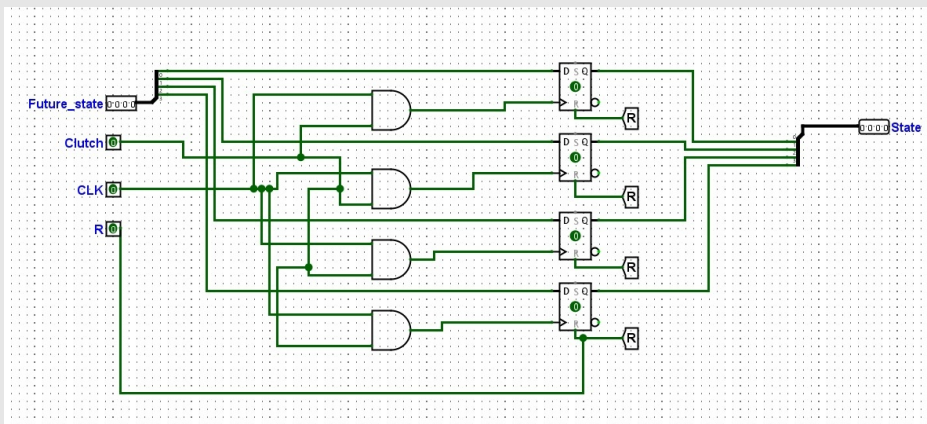
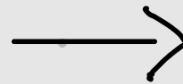
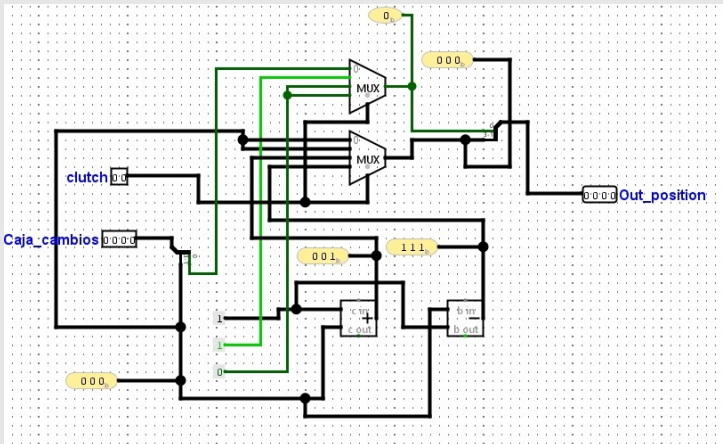
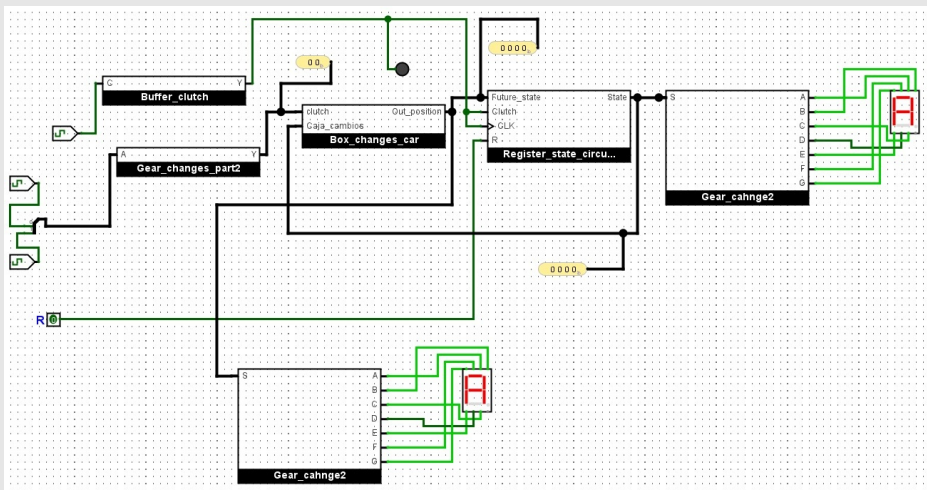


Register
State
Circuit
(next state)



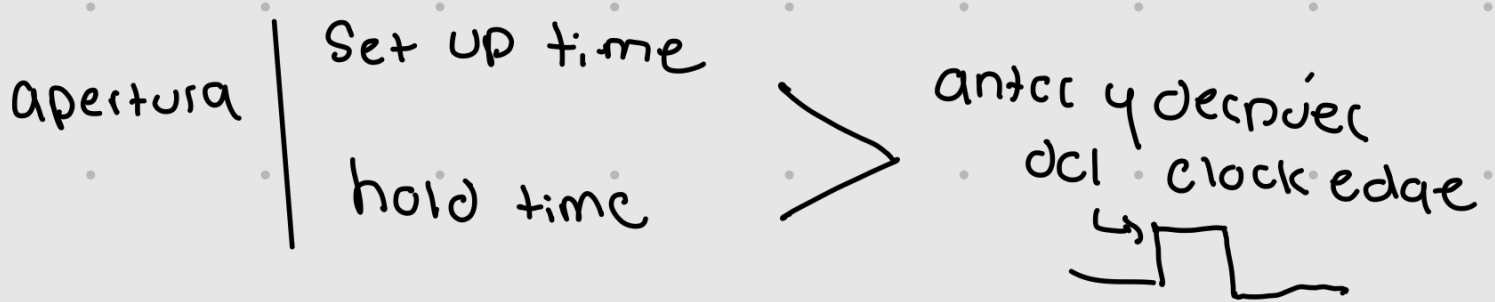
Box-changer
- car
(Current state)



FSM

¿Que pasa si D cambia al mismo tiempo que el CLK tiene su rising edge? ¿

El tiempo de apertura no permite eliminar o reducir la posibilidad de un metaestado, en ese momento el input debe ser estable para que el output sea bien definido.



- t_{ccq} = Contamination delay
- t_{pcq} = Propagation delay
- t_{setup} = Set up time
- t_{hold} = hold time

$$\text{apertura time} = t_{setup} + t_{hold}$$

- T_c = tiempo de ciclo
- $f_c = 1/T_c$ = frecuencia de clock

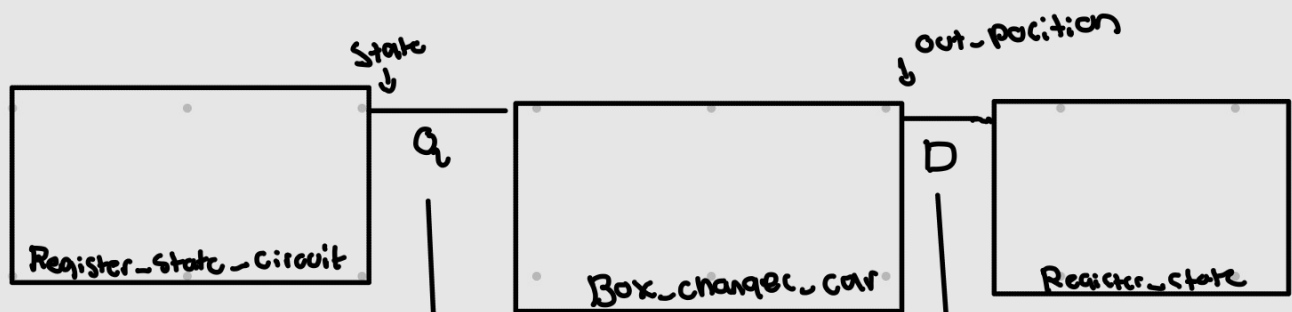
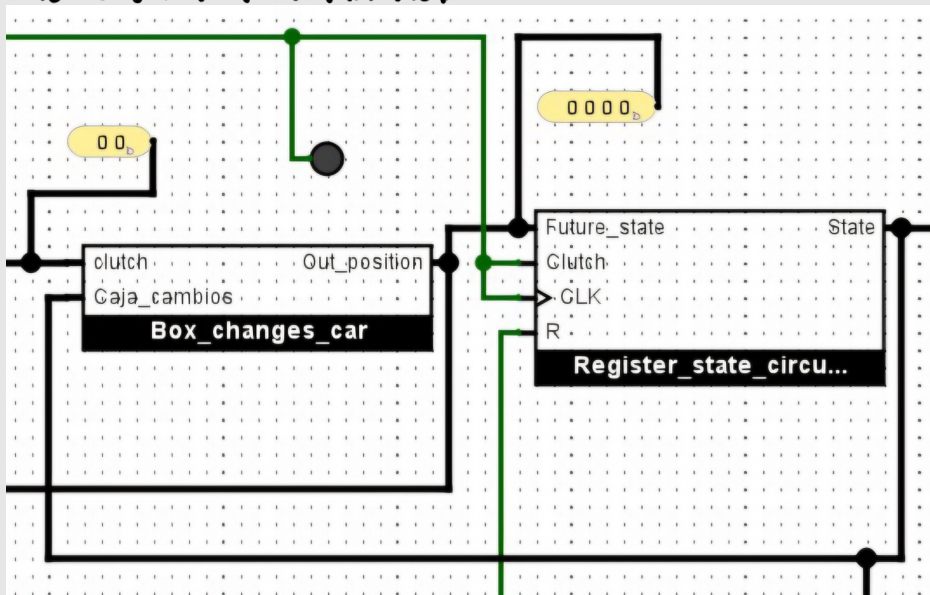
minimum clock period

$$T_c \geq t_{pcq} + t_{pd} + t_{setup}$$

max-delay constrain | Set up time constrain

$$t_{pd} \leq T_c - (t_{pcq} + t_{setup})$$

Tramo FlipFlop a FlipFlop



Flip Flop envia
su estado Q
Futuro en base
al D actual

la señal de
 Q se procesa
en los multiplexores
para priorizar
un input en base al
sumador y rector
que le envía
al Flip Flop para almacenar
una de las 14 posiciones
de la caja de cambios

MC74HC74ADTR2G

Flip-flop
Dual



$$t_{pd} = (t_{pLH}, t_{pHL})$$

$$t_{pd} = 24 \text{ ns}$$

$$t_{su} = 16 \text{ ns}$$

$$t_h = 3 \text{ ns}$$

AC ELECTRICAL CHARACTERISTICS (MC74HCT74A)

Symbol	Parameter	Guaranteed Limit			Unit
		-55 to 25 °C	≤ 85 °C	≤ 125 °C	
f_{max}	Maximum Clock Frequency (50% Duty Cycle) (Figure 4)	30	24	20	MHz
t_{pLH}, t_{pHL}	Maximum Propagation Delay, Clock to Q or $\bar{Q}$ (Figure 4)	24	30	36	ns
t_{pLH}, t_{pHL}	Maximum Propagation Delay, Set or Reset to Q or $\bar{Q}$ (Figure 5)	24	30	36	ns
t_{TLH}, t_{THL}	Maximum Output Transition Time, Any Output (Figure 4)	15	19	22	ns
C_{in}	Maximum Input Capacitance	10	10	10	pF

C _{PD}	Power Dissipation Capacitance (Per Enabled Output)*	Typical @ 25 °C, V _{CC} = 5.0 V	pF
		32	

5. Used to determine the no-load dynamic power consumption: $P_D = C_{PD} V_{CC}^2 f + I_{CC} V_{CC}$.

TIMING REQUIREMENTS (MC74HCT74A)

Symbol	Parameter	Fig.	Guaranteed Limit						Units
			-55 to 25 °C		≤ 85 °C		≤ 125 °C		
			Min	Max	Min	Max	Min	Max	
t _{su}	Minimum Setup Time, Data to Clock	6	15		19		22		ns
t _h	Minimum Hold Time, Clock to Data	6	3		3		3		ns
t _{rec}	Minimum Recovery Time, Set or Reset Inactive to Clock	5	6		8		9		ns
t _w	Minimum Pulse Width, Clock	4	15		19		22		ns
t _w	Minimum Pulse Width, Set or Reset	5	15		19		22		ns
t _r , t _f	Maximum Input Rise and Fall Times	4		500		500		500	ns

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MC74HC74A, MC74HCT74A

TIMING REQUIREMENTS (MC74HC74A)

Symbol	Parameter	V_{CC} V	Guaranteed Limit			Unit
			-55 to 25 °C	≤ 85 °C	≤ 125 °C	
t_{su}	Minimum Setup Time, Data to Clock (Figure 6)	2.0	80	100	120	ns
		3.0	35	45	55	
		4.5	16	20	24	
		6.0	14	17	20	
t_h	Minimum Hold Time, Clock to Data (Figure 6)	2.0	3.0	3.0	3.0	ns
		3.0	3.0	3.0	3.0	
		4.5	3.0	3.0	3.0	
		6.0	3.0	3.0	3.0	
t_{rec}	Minimum Recovery Time, Set or Reset Inactive to Clock (Figure 5)	2.0	8.0	8.0	8.0	ns
		3.0	8.0	8.0	8.0	
		4.5	8.0	8.0	8.0	
		6.0	8.0	8.0	8.0	
t_w	Minimum Pulse Width, Clock (Figure 4)	2.0	60	75	90	ns
		3.0	25	30	40	
		4.5	12	15	18	
		6.0	10	13	15	
t_w	Minimum Pulse Width, Set or Reset (Figure 5)	2.0	60	75	90	ns
		3.0	25	30	40	
		4.5	12	15	18	
		6.0	10	13	15	
t_r, t_f	Maximum Input Rise and Fall Times (Figure 4)	2.0	1000	1000	1000	ns
		3.0	800	800	800	
		4.5	500	500	500	
		6.0	400	400	400	

SN74LS283N

Adder 4 bits



t_{pd} = 24 ns

electrical characteristics over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER		TEST CONDITIONS†	SN54283			SN74283			UNIT
			MIN	TYP‡	MAX	MIN	TYP‡	MAX	
V _{IH}	High-level input voltage								V
V _{IL}	Low-level input voltage			0.8			0.8		V
V _{IK}	Input clamp voltage	V _{CC} = MIN, I _I = -12 mA		-1.5			-1.5		V
V _{OH}	High-level output voltage	V _{CC} = MIN, V _{IH} = 2 V, V _{IL} = 0.8 V, I _{OH} = MAX	2.4	3.6		2.4	3.6		V
V _{OL}	Low-level output voltage	V _{CC} = MIN, V _{IH} = 2 V, V _{IL} = 0.8 V, I _{OL} = MAX		0.2 0.4			0.2 0.4		V
I _I	Input current at maximum input voltage	V _{CC} = MAX, V _I = 5.5 V			1			1	mA
I _{IH}	High-level input current	V _{CC} = MAX, V _I = 2.4 V			40			40	µA
I _{IL}	Low-level input current	V _{CC} = MAX, V _I = 0.4 V			-1.6			-1.6	mA
I _{OS}	Short-circuit output current §	Any output except C4 Output C4	V _{CC} = MAX	-20	-55	-18	-55		
	-20			-70	-18	-70			
I _{CC}	Supply current	V _{CC} = MAX, Outputs open	All 8 low, other inputs at 4.5 V			56			mA
			All inputs at 4.5 V			66 99			

†For conditions shown as MIN or MAX, use the appropriate value specified under recommended operating conditions.
‡All typical values are at V_{CC} = 5 V, T_A = 25° C.
§Only one output should be shorted at a time.

switching characteristics, V_{CC} = 5 V, T_A = 25° C

PARAMETER*	FROM (INPUT)	TO (OUTPUT)	TEST CONDITIONS	MIN	TYP	MAX	UNIT
t _{PLH}	C0	Any Σ _i	C _L = 15 pF, R _L = 400 Ω, See Note 3	14 21		ns	
t _{PHL}				12 21			
t _{PLH}	A _i or B _i	Σ _j		16 24		ns	
t _{PHL}				16 24			
t _{PLH}	C0	C4	C _L = 15 pF, R _L = 780 Ω, See Note 3	9 14		ns	
t _{PHL}				11 16			
t _{PLH}	A _i or B _i	C4		9 14		ns	
t _{PHL}				11 16			

*t_{PLH} = propagation delay time, low-to-high-level output
*t_{PHL} = propagation delay time, high-to-low-level output
NOTE 3: Load circuits and voltage waveforms are shown in Section 1.

SN74HC153PWR

Multiplexor



$t_{pd} = 28\text{ ns}$

5.4 Electrical Characteristics

over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER	TEST CONDITIONS ⁽¹⁾	V _{CC} (V)	T _A = 25°C			SN54HC153		SN74HC153		UNIT
			MIN	TYP	MAX	MIN	MAX	MIN	MAX	
V _{OH}	I _{OH} = −20 μA	2	1.9	1.998		1.9		1.9		V
		4.5	4.4	4.499		4.4		4.4		
		6	5.9	5.999		5.9		5.9		
	I _{OH} = −6 mA	4.5	3.98	4.3		3.7		3.84		
		6	5.48	5.8		5.2		5.34		
V _{OL}	I _{OL} = 20 μA	2		0.002	0.1		0.1		0.1	V
		4.5		0.001	0.1		0.1		0.1	
		6		0.001	0.1		0.1		0.1	
	I _{OL} = 6 mA	4.5		0.17	0.26		0.4		0.33	
		6		0.15	0.26		0.4		0.33	
I _I	V _I = V _{CC} or 0	6		±0.1	±100		±1000		±1000	nA
I _{CC}	V _I = V _{CC} or 0, I _O = 0	6			8		160		80	μA
C _i		2 to 6		3	10		10		10	pF

(1) V_I = V_{IH} or V_{IL}, unless otherwise noted.

5.5 Switching Characteristics

over recommended operating free-air temperature range, C_L = 50 pF (unless otherwise noted) (see Figure 6)

PARAMETER	FROM (INPUT)	TO (OUTPUT)	V _{CC} (V)	T _A = 25°C			SN54HC153		SN74HC153		UNIT
				MIN	TYP	MAX	MIN	MAX	MIN	MAX	
t _{pd}	A or B	Y	2		90	150		225		190	ns
			4.5		21	30		45		38	
			6		17	26		38		32	
	Data (Any C)	Y	2		73	126		189		158	
			4.5		17	28		42		35	
			6		14	23		35		29	
	G	Y	2		38	95		150		125	
			4.5		11	19		28		24	
			6		9	16		24		20	
t _l		Y	2		20	60		90		75	ns
			4.5		8	12		18		15	
			6		6	10		15		13	

$$t_{pdC} = t_{pd}(\text{adder}) + t_{pd}(\text{multiplexer})$$

$$t_{pd} = 24 \text{ ns} + 28 \text{ ns}$$

$$t_{pd} = 52 \text{ ns}$$

minimum clock Period

$$T_c \geq t_{pcq} + t_{pd} + t_{su}$$

$$T_c = 24 \text{ ns} + 52 \text{ ns} + 16 \text{ ns}$$

$$T_c = 92 \text{ ns}$$

$$f_c = 1/T_c = 1/92 \text{ ns}$$

$$f_c = 10.86 \text{ MHz}$$

$$t_{pd} = 92 \text{ ns} - (24 \text{ ns} + 16 \text{ ns})$$

$$t_{pd} = 92 \text{ ns} - (40 \text{ ns})$$

$$t_{pd} = 52 \text{ ns}$$

hold time

$$t_{ccq} + t_{cd} \geq t_{hold}$$

$$t_{cd} \geq t_{hold} - t_{ccq}$$

$$t_{cd} = 5.6 \text{ ns} + 4.9 \text{ ns} = 10.5 \text{ ns}$$

$$t_{ccp} = 5.25 \text{ ns}$$

$$5.25 \text{ ns} + 10.5 \text{ ns} \geq 3 \text{ ns}$$

$$15.75 \geq 3 \text{ ns}$$

$$10.5 \text{ ns} \geq 3 \text{ ns} - 5.25 \text{ ns}$$

$$10.5 \text{ ns} \geq -2.25 \text{ ns}$$